

Chapter 14: The "Brain" (Databases)

I. The Amnesia Problem & Relational Databases (SQL)

- **The Problem:** Without a database, every dynamic website has _____; it forgets you the second you close the tab.
- **Database Purpose:** Solves the two biggest problems: provides _____ (permanently stores data) and _____ (structured retrieval).

Relational Databases (SQL)

- **Structure:** Organizes data into _____ that look like spreadsheets.
- **Language:** To talk to this type of database, you use _____ (SQL).
- **Key Components:**
 - **Primary Key:** A column (usually id) that _____ identifies every row in a table.
 - **Foreign Key:** A column in one table that _____ to the primary key in another table, creating a relationship.

Deep Dive Discussion

1. **Memory Problem:** You build a social media site that allows users to create posts. If you don't use a database, where does that user's post exist, and what happens to it when the user refreshes the page?
2. **Relational Magic:** Explain how a relational database links a specific high score (e.g., 998,000) to a specific user name (e.g., ArcadeAce) using the **Primary Key** and **Foreign Key** concept.

Application Activity

1. **Tracing the Key:** Imagine the HighScores table has a score with user_id = 1. In the Users table, circle the **Primary Key** that the score is referencing and the **Foreign Key** that makes the connection.

II. Database Types & SQL Commands

Comparing Database Types

Database Type	Analogy	Description	Use Case
SQL (Relational)	_____	Strict tables and	Accounts, Orders,

	or Filing Cabinet	columns; excellent for complex, _____ data.	_____ (core site data).
NoSQL (Non-Relational)	_____ of JSON Files	Flexible; stores data in JSON-like _____.	Live Chat messages, rapidly _____ data (like "likes").
Local Storage	_____ on the Browser	Tiny storage (5-10MB); data is _____ (does not go to a server).	Dark Mode preference, last _____ term.

The Language of the Database (CRUD)

CRUD	SQL Command	Purpose
C reate	_____	Adds a new row of data (e.g., signing up).
R ead	_____	_____ existing data (e.g., loading a gallery).
U pdate	_____	Modifies an existing row (e.g., changing a password).
D elete	_____	Removes a row (e.g., deleting a post).

Deep Dive Discussion

1. **Local vs. External:** Why did the book suggest using **Local Storage** for a Dark Mode toggle, but you **must** use an external SQL database for "Competitive Cody's" high score?
2. **SQL Query:** In the SELECT query structure (SELECT columns FROM table WHERE

condition), what is the purpose of the WHERE clause?

Application Activity

1. **SQL Query Construction:** Competitive Cody (`user_id` 1) logs in. You need to pull his two highest scores, regardless of the game. Fill in the blanks below to write the SQL query that achieves this goal:
- SELECT `game_title`, `score_value`
 - FROM ____________________
 - WHERE `user_id` = ____________________
 - ORDER BY `score_value` DESC
 - LIMIT 2;

III. Database Security and Server-Side Code

- **Security Rule:** Database interaction **must** happen on the _____ to prevent direct user attacks.
- **SQL Injection:** A severe security attack where a hacker inserts _____ commands into an input field to steal or damage data.
- **The Fix:** You must always _____ user input before putting it into a database query.

Password Security

- **The Problem:** Never store passwords as _____.
- **Hashing:** A _____ algorithm that scrambles a password into an irreversible " _____ " for secure storage.
- **Verification:** The system takes the user's login password, _____ it, and checks if the two scrambled strings match.

Deep Dive Discussion

1. **Hacker Protection:** A hacker manages to steal the database containing all the users' passwords. Why are the users safe if the developer used **hashing**, but they would be completely compromised if the developer stored the passwords as plain text?
2. **The Server Wall:** Why can't a developer just use JavaScript (client-side) to send an INSERT command directly to an external database?

Application Activity

1. **Security Threat:** The hacker input `Cody'; DROP TABLE Users; --` successfully executes an **SQL Injection**. Which part of that input is the actual malicious command that destroys the database?
 - **Malicious Command:**